Greedy methods exercises

A06-A01 Endpoints and cleanup

Are [1,4) and [4,6) compatible? Repeat with cleanup time δ=1 and state the general test.

A06-A02 Trace all eight activities

Use activities A through H from the slides. Sort by finish time and ID. Record L before each decision, the decision and its reason, and the output.

A06-A03 Complete the exchange proof

A proof says only, "g finishes earlier than o, so replace o." Supply arguments for the fixed prefix, every later activity, and unchanged count. Is the conclusion existential or unique?

A06-A04 Refute two proposed rules

Give counterexamples with a strict first choice for earliest start and shortest duration. The latter repeatedly picks a shortest activity compatible with every selected activity. State each output of each rule and an optimum.

A06-A05 Handle empty input and ties

What fails when the answer is initialized with the first activity? Explain why finish-time ties can be broken arbitrarily. State the costs for sorted and unsorted input.

A06-A06 Weighted recurrence and traceback

A=(0,2,2), B=(2,4,2), C=(0,4,7), with fields start, finish, weight. Break finish-time ties by ID. Compute p, M, and an optimal set. Explain the failure of earliest finish.

A06-H01 Prefix-free and Kraft

Is {0,01,10} prefix-free or uniquely decodable? Give a counterexample. Can lengths 1,2,3,3 form a binary prefix-free code? Construct one and prove that lengths 1,1,2 are impossible.

A06-H02 Reconstruct the six-symbol example

Use a 45,b 13,c 12,d 16,e 9,f 5. Record both extracted items, the new node, and the queue after every round. Give the root weight and codewords. Verify the cost with weighted leaf depths and merge weights separately.

A06-H03 The exchange inequality

Minimum symbol x has depth 2 and weight 3. A deepest leaf a has depth 5 and weight 8. Compute the cost change of swapping them. Explain how to place the second minimum symbol in the sibling position and why existence of one compatible optimal tree suffices.

A06-H04 Contraction and a wrong greedy rule

Prove B(T)=B(T′)+w_x+w_y when contracting two sibling leaves. For weights 1,2,3,4, test the rule that repeatedly merges the two largest weights. List merge costs for both strategies.

A06-H05 Ties and boundary cases

For four symbols of weight 1, give two different optimal codebooks and their costs. Handle one symbol of weight 7, weights 0,0,5, and an all-zero alphabet. Distinguish empty-word codes from a format requiring at least one bit per codeword.

A06-H06 Tree construction versus codeword output

Explain heap construction time and space for unsorted weights. How do two queues give linear construction for sorted weights? Why is O(n) space not always enough to materialize all codeword strings?

A06-M01 | Axioms and bases

Let $E=\{a,b,c,d\}$ and $\mathcal{I}=\{X\subseteq E:|X\cap\{a,b\}|\le 1 \text{ and } |X\cap\{c,d\}|\le 1\}$. Prove this is a matroid. List all bases and the rank.

A06-M02 | Activity exchange failure

For a=[0,4), b=[0,2), c=[2,4), list all independent sets and a pair that violates exchange. Set w(a)=5 and w(b)=w(c)=3. Trace decreasing-weight greedy and find the optimum.

A06-M03 | Negative weights and the objective

In $U_{2,3}$, let w(a)=8, w(b)=−1, w(c)=−6. Find a maximum-weight independent set and a maximum-weight basis. Explain why adding 7 to all weights may change the former but cannot change the ranking of bases.

A06-M04 | A forest and one exchange

Trace greedy on the five weighted edges in the slides. Then take G={AB}, g=BC, and basis O={AB,AC,CD}. Write each extension of H, the removed element y, and the weight change.

A06-M05 | Changed penalties

Keep the seven deadlines but replace p_i with 80−p_i. Give the full selection trace in decreasing new penalty order. Supply an N_t>t witness for each rejection. Give a complete optimal schedule and its total penalty.

A06-M06 | Exchange and the model boundary

Prove exchange for unit jobs using N_t. Then allow unequal processing times. Job a takes 2 time units and has deadline 2. Jobs b,c each take 1 time unit and have deadline 2. Is this still a matroid? Identify the failed axiom.

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd ed., Chapter 16, Greedy Algorithms. Original handwritten scanned algorithms notes (演算法_wjungle), greedy-method pages covering activity selection, Huffman coding, matroids, and scheduling. MIT 6.046J, Fall 2004, Lecture 13, Greedy Algorithms and Minimum Spanning Trees; Fall 2005, Lecture 16. These MIT lectures support the greedy and MST proof framework, not the original Huffman exposition.

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https://github.com/xinhi6716/computer-science-self-study